IDENTITY THREAT SCREENING

PersonaTrace AI

AI-Based Deepfake Interview Risk Detection System for Remote Hiring

A practical computer vision and cybersecurity project that analyzes interview videos for face consistency, behavior patterns, speech-mouth timing, and video artifact clues.

Full Project Explanation and Technical Report

1. Project Overview

PersonaTrace AI is a remote interview risk-screening system. A user uploads a short interview video, and the system analyzes visual, behavioral, audio-video, and quality signals to produce an explainable interview risk score. The goal is not to prove that a person is fake, but to identify suspicious evidence that should trigger manual review.

Core idea: remote hiring now has a trust problem. A candidate on camera could be using a deepfake face, proxy speaker, virtual camera, face-swap tool, or mismatched identity. PersonaTrace AI gives recruiters or security reviewers a structured way to inspect risk.

2. Problem It Solves

Remote interviews are convenient, but they reduce identity assurance. Companies may make hiring decisions, grant system access, or continue onboarding based on a video call that could be manipulated. This can lead to wrong hiring, identity fraud, data exposure, and security risk.

- Detects missing or unstable face presence across the interview clip.
- Flags suspicious behavior such as very low blink variation or frozen face motion.
- Checks whether speech activity appears without matching mouth movement.
- Highlights video quality clues such as blur, unstable lighting, and abrupt frame jumps.
- Produces a clear score and reasons instead of a black-box yes/no decision.

3. User Workflow

Opening Screen	Animated PersonaTrace AI entry page introduces the identity threat screening workflow.
Evidence Upload	User uploads a short interview video in MP4, MOV, AVI, or MKV format.
Signal Extraction	The system extracts metadata, sampled frames, audio, face regions, mouth regions, and quality metrics.
Trace Report	The app shows an interview risk score, risk level, key findings, and module-level metrics.

4. System Modules

Module	What It Checks	Output
Video Intake	Reads uploaded video, FPS, frame count, duration, resolution, and sampled frames.	Video metadata and evidence frames
Face Consistency	Detects face presence, multiple faces, face position jitter, and face size jitter.	Face risk score and reasons
Behavior Pattern	Checks eye signal, blink-like state changes, and frozen/static face motion.	Behavior risk score
Lip-Sync Signal	Extracts audio, estimates speech activity, checks mouth movement, and scores speech-mouth mismatch.	Lip-sync risk score
Video Quality	Checks blur, lighting variation, and abrupt frame differences.	Quality risk score
Risk Report	Combines module scores into a final explainable interview risk score.	Final score, level, findings

5. How The Detection Works

Face consistency

OpenCV face detection is applied to sampled frames. PersonaTrace AI measures whether a face appears consistently, whether multiple faces appear, and whether face position or size changes sharply across frames.

Blink and frozen face behavior

The system looks for eye-region signals and face crop changes over time. A face with very low movement or no blink-like variation can be flagged as suspicious or unnatural, especially in longer clips.

Speech and mouth timing

Audio is extracted with imageio-ffmpeg. The system estimates speech activity using audio energy windows and compares it with mouth-region motion between frames. If speech exists while mouth movement is weak, the lip-sync risk increases.

Video quality artifacts

The quality module calculates frame sharpness using Laplacian variance, lighting variation using brightness changes, and possible glitches using abrupt frame differences.

6. Risk Scoring

The final interview risk score is a weighted combination of module scores. If audio is available, face consistency, behavior, lip-sync, and video quality all contribute. If audio is unavailable, the system still produces a visual risk score from face, behavior, and quality.

- Low risk: no major warning signals in sampled windows.
- Medium risk: one or more signals need human review.
- High risk: multiple warning signals suggest possible manipulation or identity risk.

Important: the score is a screening signal, not a final identity verdict. High-risk results should trigger manual verification, additional identity checks, and review by a human.

7. Technology Stack

Python	Core programming language for video processing and scoring logic.
Streamlit	Interactive web app interface with intro, upload, and report pages.
OpenCV	Video reading, frame extraction, face detection, image differences, blur checks.
NumPy	Numerical processing for frame, audio, and scoring calculations.
imageio-ffmpeg	Bundled FFmpeg helper for audio extraction from uploaded video files.

8. Project File Structure

app.py controls the Streamlit UI and app flow. The src/personatrace package contains separate modules for video analysis, face consistency, behavior analysis, lip-sync analysis, video quality analysis, and final report scoring.

- app.py - Streamlit interface and user workflow.
- video_analysis.py - metadata and frame extraction.
- face_analysis.py - face detection and consistency risk.
- behavior_analysis.py - blink and frozen face behavior.
- lip_sync_analysis.py - audio-video speech-mouth timing.
- video_quality_analysis.py - blur, lighting, and glitch checks.
- report.py - final risk score and findings.

9. Limitations

- It is not a forensic-grade detector and should not be treated as proof of fraud.
- Accuracy depends on video quality, lighting, camera angle, face visibility, and audio availability.
- OpenCV-based face and eye detection can miss faces in difficult angles or low-light scenes.
- Lip-sync scoring is approximate because it uses sampled windows rather than a trained lip-reading model.
- The best use is screening and explanation, followed by human review.

10. Resume And Demo Value

PersonaTrace AI is a strong resume project because it combines AI, computer vision, cybersecurity, audio-video analysis, and a real remote hiring problem. It is more unique than a standard chatbot or resume analyzer because it addresses modern identity fraud risk.

Resume line: Built PersonaTrace AI, a Streamlit and OpenCV-based deepfake interview risk screening system that analyzes face consistency, blink and motion behavior, speech-mouth timing, and video artifacts to generate an explainable remote hiring risk score.